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hw2_me501_Q2b.m

(C) Andrew Sabelhaus, 2026

```
clear all;
close all;
clc;
```

```
disp('ME 501 HW2 simulation of pneumatic chambers dynamics, comparison with
exact solution.')
```

ME 501 HW2 simulation of pneumatic chambers dynamics, comparison with exact solution.

Setup: constants and initial conditions

```
tmax = 5;
dt = 0.05;
```

```
n = tmax/dt; % number of timesteps, for loop-ing
% somehow matlab doesn't round correctly. sometimes...
n = round(n);
```

```
% The matrix for our linear dynamics is
A = [-3  1  0;
      1 -2  1;
      0  1 -3];
```

```
%%%% Initial conditions:
```

```
p1_0 = 5;
p2_0 = 0;
p3_0 = 50;
x0 = [p1_0; p2_0; p3_0];
```

Simulate to obtain trajectory - Numerical, approximated!

```
% insert our initial condition as the 1st element in traj
% note that traj will have n+1 columns.
```

```

x_traj = zeros(size(x0,1), n+1);
x_traj(:,1) = x0;

% actual simulation! Use Forward Euler for now.
for t=1:n
    %%%% DYNAMICS
    bolddotx_t = A*x_traj(:,t);
    % Forward Euler
    x_traj(:,t+1) = x_traj(:,t) + dt*bolddotx_t;
end

```

Exact Solution for comparison

```

% let's precalculate for a one-to-one comparison
x_traj_exact = zeros(size(x0,1), n+1);
x_traj_exact(:,1) = x0;

% precalculate via iteration
for k=1:n
    % The exact solution is  $x(t) = e^{(A)t}x_0$ 
    % Since "t" is inside the exponential, need to re-call this function
    % each iteration.
    t = k*dt; % here, k is index, dt is timestep, so total time is t.
    %%%% If exponentiating at each timestep:
    Phi_t = expm(A*t);
    x_traj_exact(:,k+1) = Phi_t*x0;
end

```

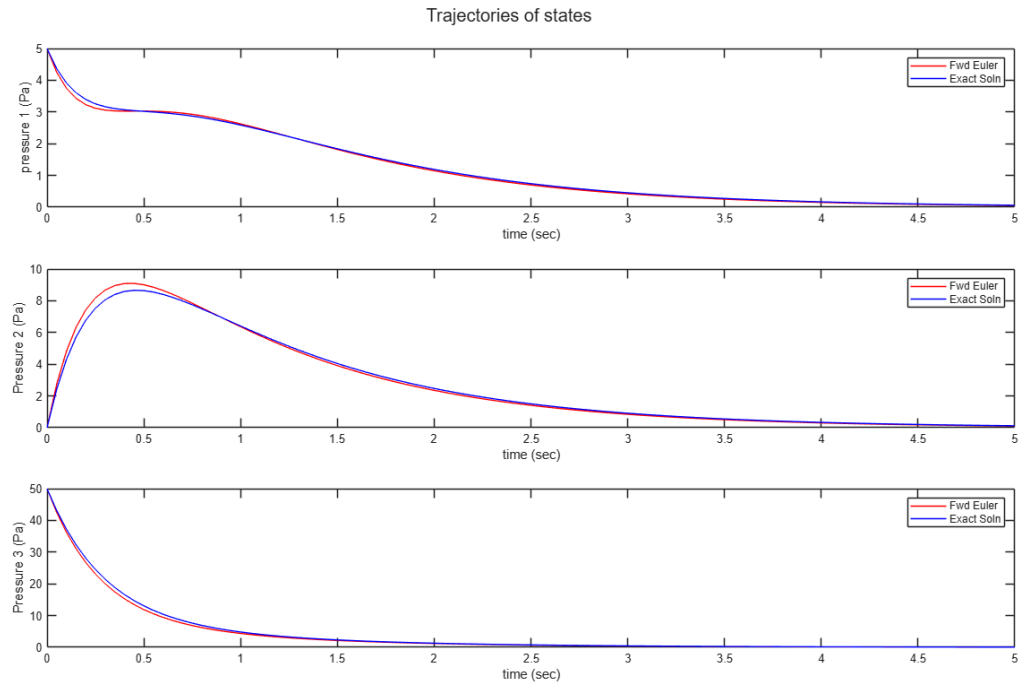
Plot

```

% We can also plot the three time series trajectories of theta, dottheta,
% and u
figure;
hold on;
t = [0:dt:tmax];
subplot(3,1,1);
sgtitle('Trajectories of states');
% pressure 1
plot(t, x_traj(1,:), 'r');
hold on;
plot(t, x_traj_exact(1,:), 'b');
legend(["Fwd Euler", "Exact Soln"]);
xlabel('time (sec)');
ylabel('pressure 1 (Pa)');
% pressure 2
subplot(3,1,2);
plot(t, x_traj(2,:), 'r');
hold on;
plot(t, x_traj_exact(2,:), 'b');
legend(["Fwd Euler", "Exact Soln"]);
xlabel('time (sec)');
ylabel('Pressure 2 (Pa)');
% pressure 3

```

```
subplot(3,1,3);
plot(t, x_traj(3,:), 'r');
hold on;
plot(t, x_traj_exact(3,:), 'b')
legend(["Fwd Euler", "Exact Soln"]);
xlabel('time (sec)');
ylabel('Pressure 3 (Pa)');
```



Published with MATLAB® R2026a